جامعة مصر للمعلوماتية
EGYPT UNIVERSITY
OF INFORMATICS

Egypt University of Informatics
Computer and Information Systems
Data Analysis Course

The Analysis of the Performance of Data Analysis Students

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1.Introduction

Procrastination the act of delaying tasks despite knowing the consequences is one of the most common behavioral patterns among students and working professionals. In an increasingly distracted world dominated by social media, short-form content, and constant notifications, the ability to start and complete tasks on time has become a genuine challenge for many.

This report investigates the relationship between procrastination habits and academic or work performance, with a particular focus on how delay patterns, digital distractions, and time management practices collectively shape the quality of individuals' output. Understanding these dynamics is important not only for academic institutions but for workplaces seeking to improve productivity and well-being.

Research Question

To what extent does procrastination negatively affect the academic and professional performance of students and employees, and what are the primary drivers of procrastination behavior?

Hypothesis

Individuals who frequently procrastinate are more likely to report lower work quality and reduced academic or work performance.

2.Methodology

Population of Interest:

The target population for this study is students and employees aged 18 and above, with a primary focus on university students aged 18–25, as they represent the demographic most commonly associated with academic procrastination. The survey was distributed digitally to reach respondents in their natural environments (online communication channels), where procrastination inducing distractions are most prevalent.

Sampling Method:

A convenience sampling method was used for this study. The survey link was shared through digital platforms such as WhatsApp groups & social media. This approach was chosen for its practicality and speed, allowing the collection of 85 genuine responses within a short timeframe. While convenience sampling is not probabilistic, it is appropriate here given the exploratory nature of the study and the difficulty of accessing a fully random sample of the target population.

To reduce the risk of sampling exclusively from a single group, the survey was shared across multiple channels and peer networks, ensuring some degree of demographic variation.

Bias Identification:

Several potential sources of bias were identified and addressed:

- Self-reporting bias: Respondents may understate or overstate their procrastination habits due to social desirability. To mitigate this, questions were framed neutrally without implying a "correct" answer.
- Sampling bias: As this is a convenience sample, the results may over-represent university students aged 18–21. This is acknowledged as a limitation.

- Response bias: All questions were optional, and no identifiable personal information was collected, encouraging more honest responses.
- Question clarity: Each survey question was reviewed to avoid leading language (e.g., using "How often do you procrastinate?" rather than "Do you procrastinate a lot?").

Survey Questions:

- What is your age group? (18 and below / 18–21 / 21–25 / 25+)
- You are a(n): (Student / Employee)
- How often do you procrastinate? (Never / Rarely / Sometimes / Often / Always)
- When do you usually start your tasks? (Once announced / A few days after / A day before / The day of the due date)
- What are your main reasons for procrastinating? (Multi-select)
- What distracts you the most from your tasks? (Social media / TV/Movies / Games / Friends/Family)
- How many hours do you spend on your phone daily? (Less than 2 / 2–4 / 4–7 / 8+)
- How would you rate the quality of your work when completed last minute? (1–5)

Online survey link: https://docs.google.com/forms/d/e/1FAIpQLSff-z-4YnkZPCMDmBtjUJHF27hCTbddXs0AR0DDQahlxhzwEQ/viewform?usp=dialog

Number of samples collected: 85

3.Analysis

To analyze the survey data, descriptive statistics were calculated for the numeric variables (Work Quality and Academic Performance ratings). The mean, median, and mode were computed for each.

Statistics	Quality of work	Overall Academic/Work performance
Mean	3.27	3.18
Median	3	3
Mode	3	3

Figure 1: How Often Do Respondents Procrastinate?

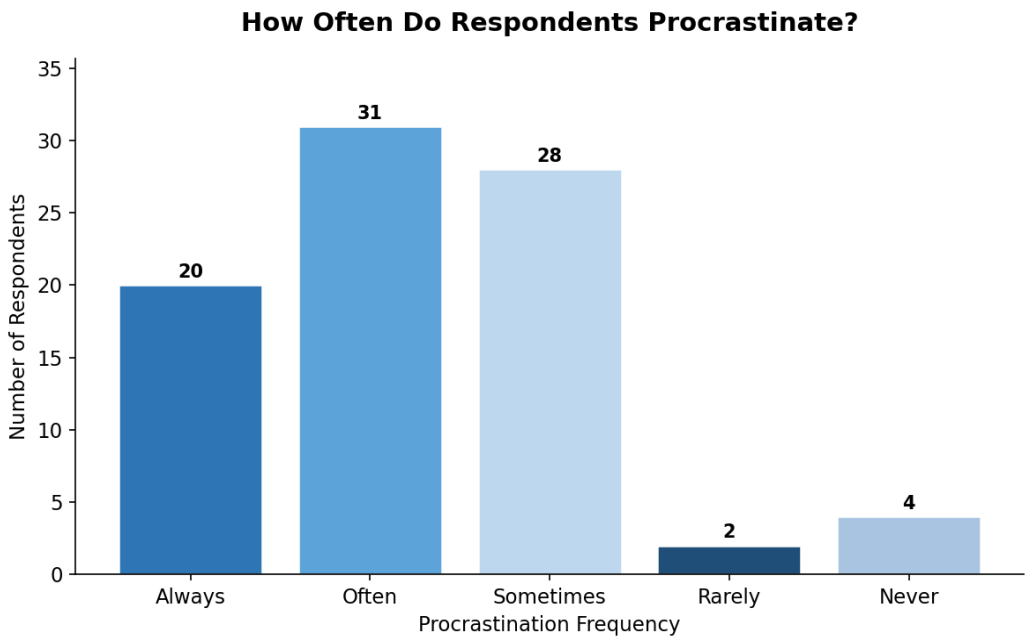

Figure 1- Distribution of self-reported procrastination frequency among 85 respondents.

The bar chart reveals that the majority of respondents identify as frequent procrastinators, with "Often" being the most common response (31 respondents, 36.5%), followed closely by "Sometimes" (28 respondents, 32.9%). A notable 20 respondents (23.5%) report procrastinating "Always", meaning that nearly 60% of the sample procrastinates often or always. Only a small minority (6) respondents combined report rarely or never procrastinating, indicating that procrastination is a widespread and dominant behavioral pattern within this sample. This distribution strongly supports the initial hypothesis that frequent procrastination is common among students and employees.

Figure 2: Task Start Timing

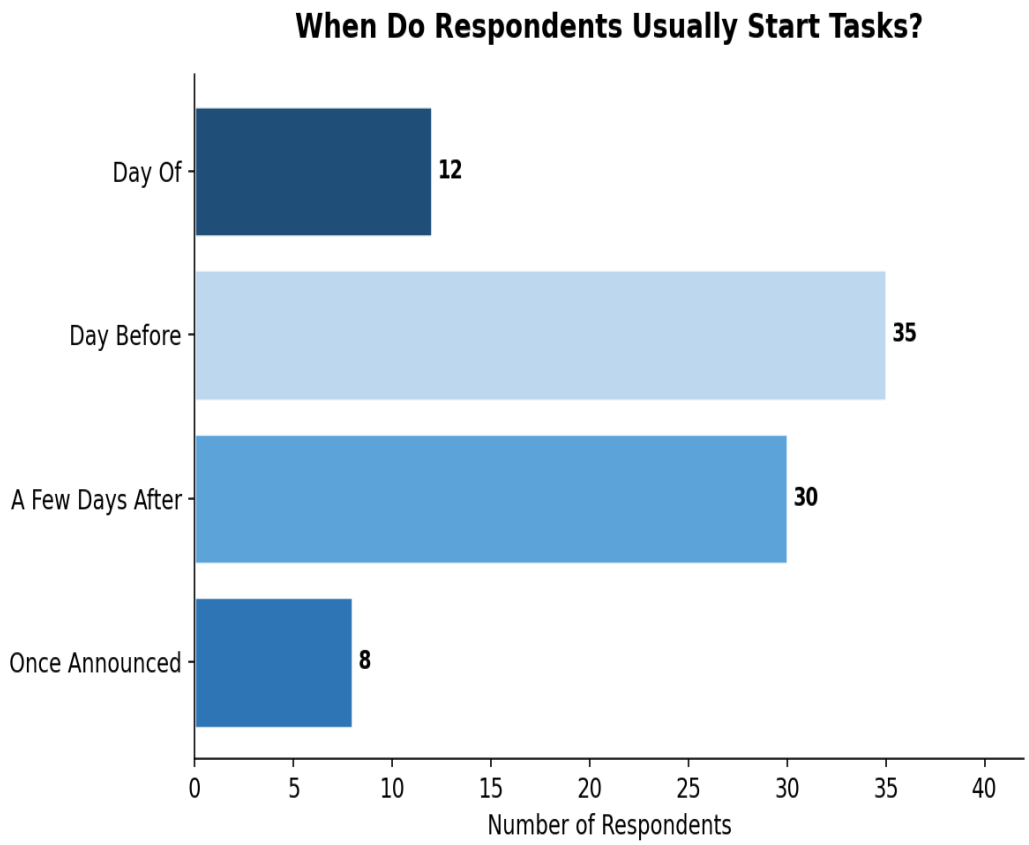

Figure 2- When respondents typically begin working on assigned tasks relative to the deadline.

The horizontal bar chart demonstrates a clear deadline-driven behavior pattern: 35 respondents (41.2%) begin their work the day before the due date, and 12 (14.1%) wait until the actual day of the deadline meaning that over half of all respondents begin their tasks within 24 hours of the deadline.Only 8 respondents (9.4%) begin immediately upon announcement. This trend confirms that procrastination does not just mean delay it specifically manifests as extreme last-minute behavior. This has direct implications for quality, as tasks started so close to deadlines leave little room for revision or reflection.

Figure 3: Reasons for Procrastinating

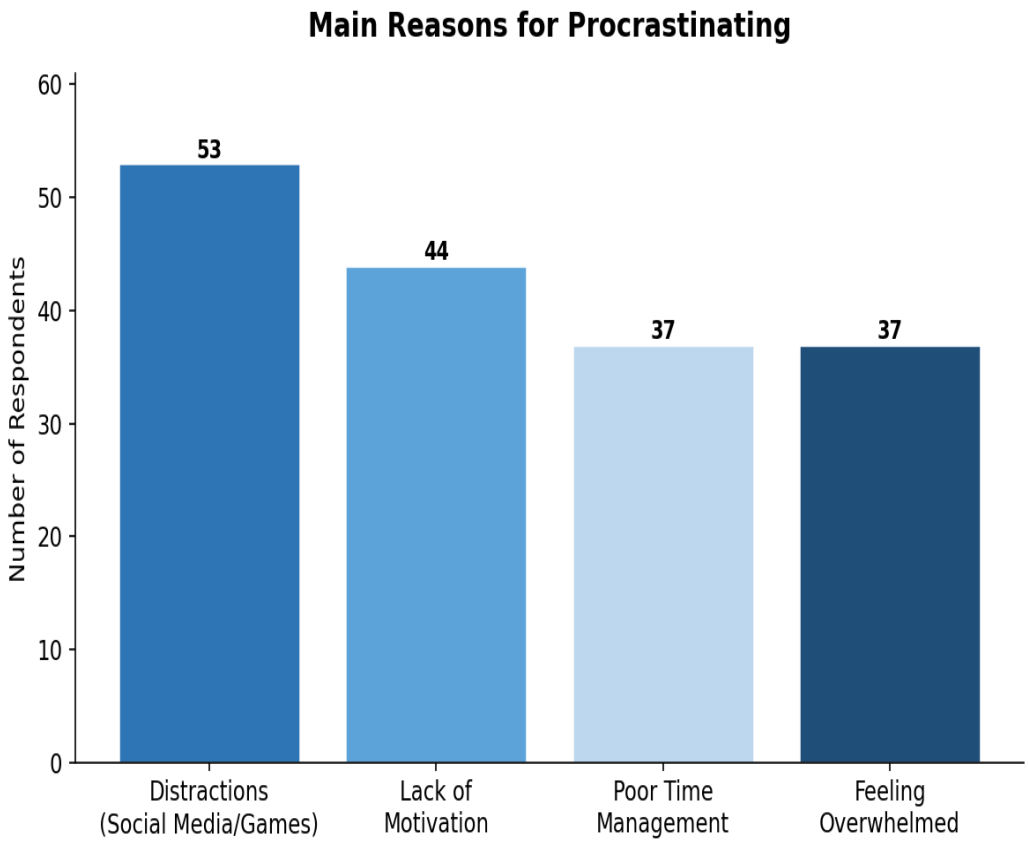

Figure 3 - Top four reasons selected by respondents for why they procrastinate (multi-select question).

Distractions from social media and games emerged as the most frequently cited reason for procrastination (53 mentions), followed by lack of motivation (44), poor time management (37), and feeling overwhelmed (37). Because this was a multi-select question, many respondents cited multiple overlapping reasons, suggesting that procrastination is rarely caused by a single factor. The dominance of distractions alongside motivational issues points to both an environmental and psychological dimension of procrastination an external pull from technology combined with an internal difficulty in engaging with tasks.

Figure 4: Self-Rated Work Quality and Performance

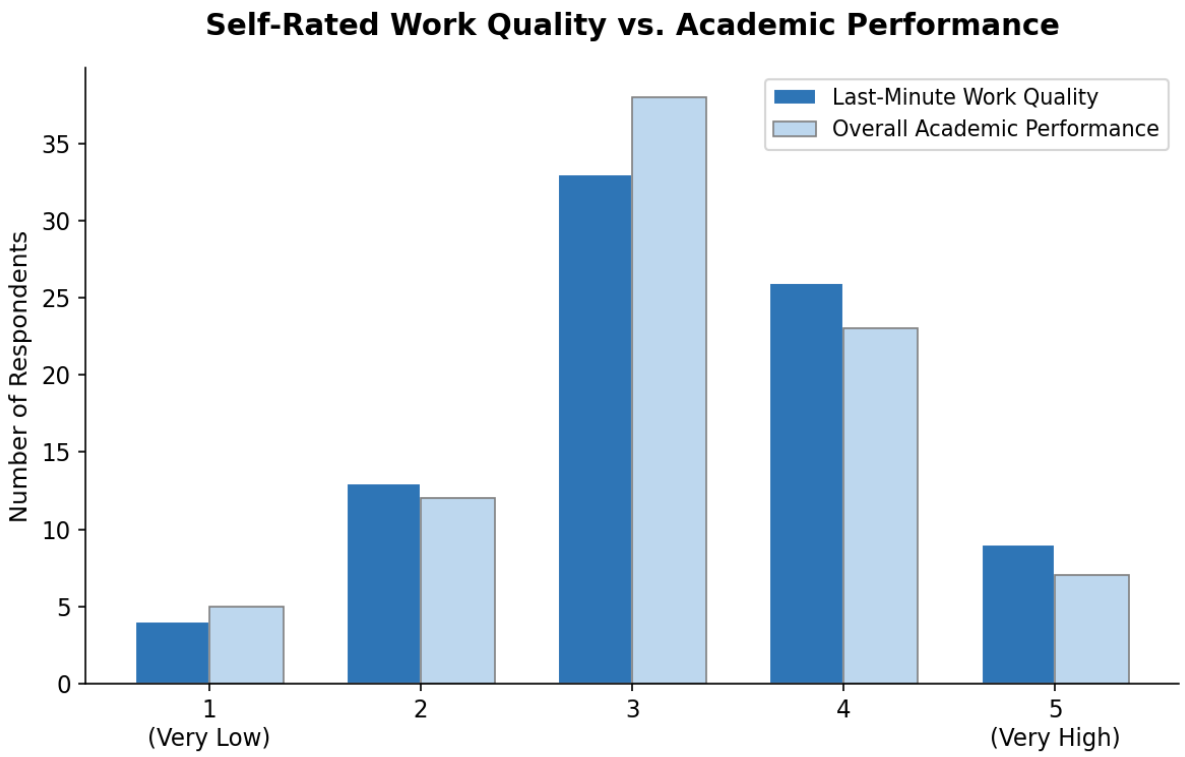

Figure 4- Distribution of respondents' self-rated last-minute work quality vs. overall academic/work performance (scale: 1–5).

Both rating distributions are centered around 3 (Mean: 3.27 for work quality, 3.18 for performance), with similar standard deviations just under 1.0. This indicates that respondents perceive their last-minute work as mediocre neither distinctly poor nor good. Interestingly, last minute work quality ratings are marginally higher on average than overall performance ratings, which may suggest that respondents feel they "get by" with last-minute work but recognize its broader cost on their sustained performance. The symmetry of both distributions around the midpoint reflects a realistic self-awareness among respondents about the moderate but real consequences of procrastination.

Figure 5: Impact on performance

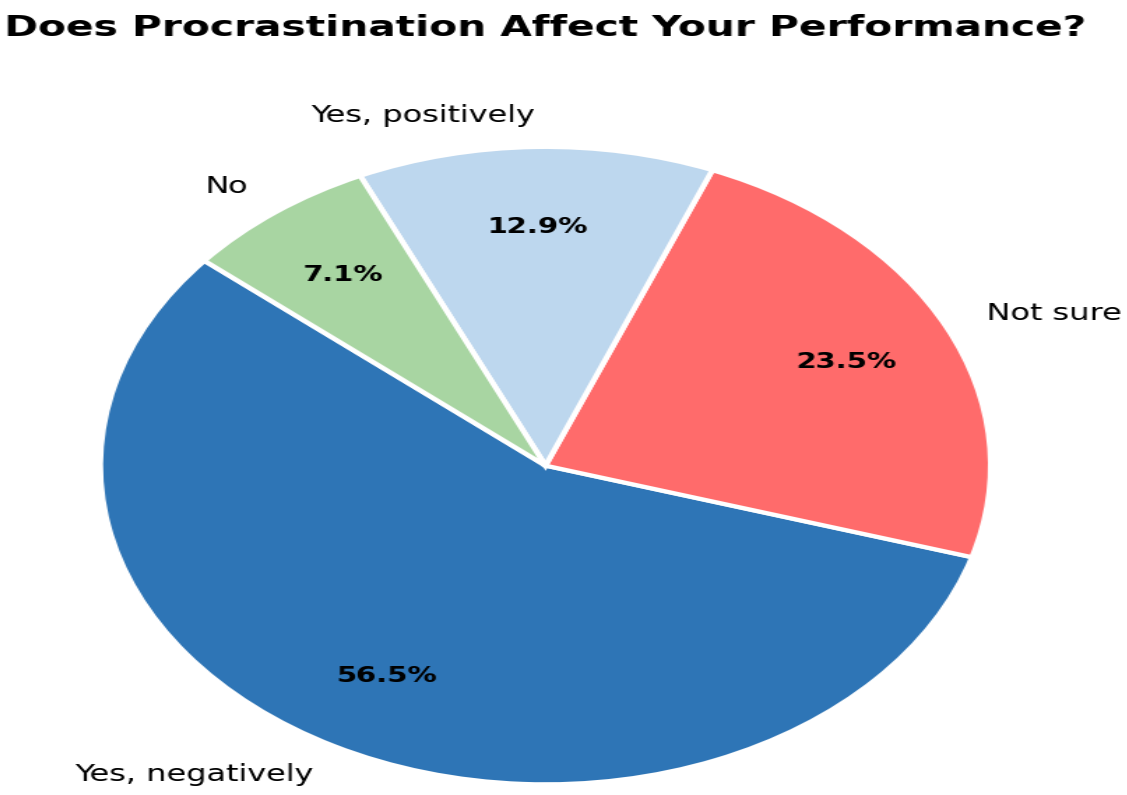

Figure 5 - Respondents' self-assessment of whether procrastination negatively affects their performance.

A clear majority of respondents 56.5% acknowledge that procrastination affects them negatively. An additional 23.5% are unsure, which may reflect ambivalence or inconsistency in their experience. Only 12.9% believe procrastination positively affects their performance (possibly through pressure induced focus), and 7.1% report no effect. The high proportion of "Yes, negatively" responses, combined with the large "Not sure" group, suggests that most respondents have either directly experienced or suspect performance consequences from their procrastination habits, making this a practically significant finding.

Figure 6: Daily phone usage

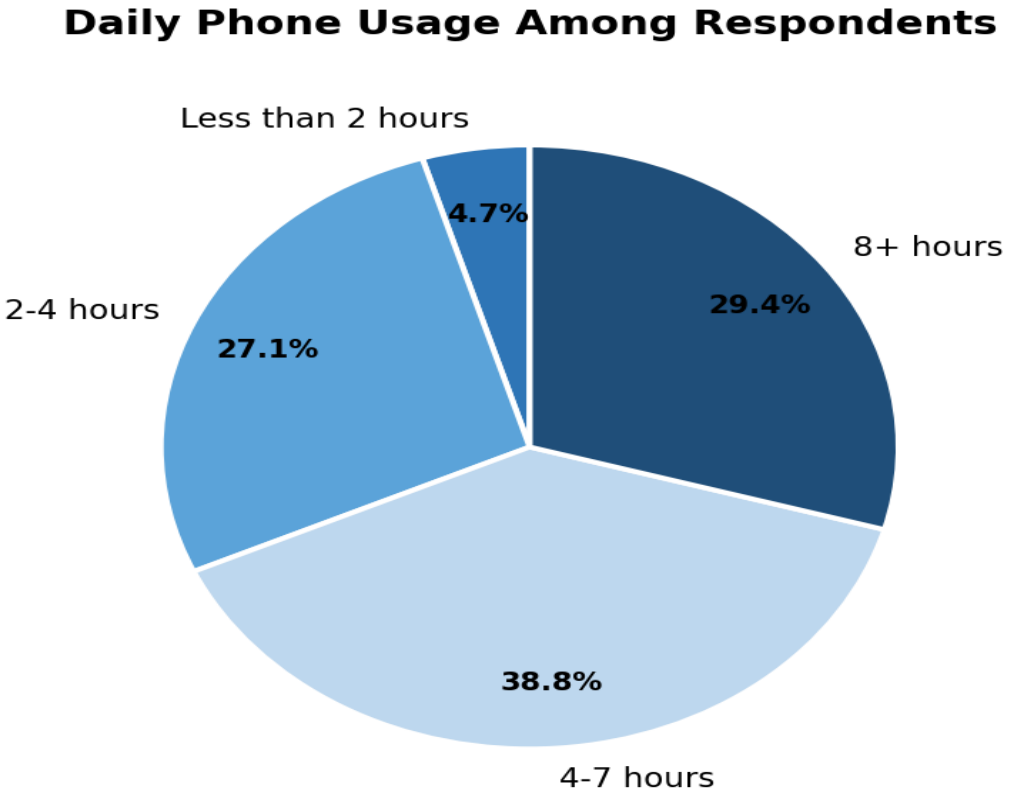

Figure 6- Distribution of respondent's self-reported daily phone usage in hours.

The pie chart shows that 38.8% of respondents spend 4–7 hours on their phone daily, and 29.4% spend 8 or more hours meaning that over two-thirds of the sample spends at least 4 hours per day on their phones. Given that social media was identified as the top distraction and reason for procrastination, this high phone usage figure provides important contextual support: the same devices that enable communication and learning are also the primary enablers of procrastination. Only 4.7% of respondents report using their phones for less than 2 hours per day.

4.Conclusion

This study aimed to examine the extent of procrastination among students and employees and explore whether it is associated with lower academic or work performance. Based on the responses collected from 85 participants, the results generally support the initial hypothesis. The majority of respondents (over 85%) reported that they procrastinate sometimes, often, or always, and more than half begin working on tasks within 24 hours of the deadline. Social media distractions and lack of motivation were the most frequently reported reasons for procrastination. In addition, the average self-rated quality of last-minute work was 3.27 out of 5, while overall performance averaged 3.18. These results suggest that respondents view their performance as moderate rather than extremely poor, even when procrastination occurs.

More importantly, 56.5% of respondents stated that procrastination negatively affects their performance, while 23.5% were unsure about its impact. This indicates that many individuals recognize that delaying tasks may influence the quality of their work and their overall productivity.

It is important to note that this study is based on self-reported responses and observational data. Therefore, it cannot establish a direct cause-and-effect relationship between procrastination and performance. In some cases, lower performance may lead to avoidance behaviors rather than the other way around. As a result, the findings should be interpreted as exploratory and correlational. Future research with larger and more diverse samples could provide deeper insights into how procrastination affects performance across different groups.

5.Critical Reflection

Several observations and considerations emerged during the data collection and analysis phases that are worth acknowledging:

- Sample composition: 78.8% of respondents were aged 18–21 and 87% identified as students. This means the results primarily reflect young student behavior and may not fully represent older professionals or non-student populations. Future studies could aim for a more balanced demographic spread to broaden the applicability of findings.
- Convenience sampling considerations: Because the survey was distributed through personal and peer networks, respondents likely share similar backgrounds and environments. This is a natural characteristic of convenience sampling and is acknowledged as a scope limitation rather than a flaw — the findings are best interpreted as representative of the immediate community surveyed.
- Question design for multi-select responses: The "reasons for procrastinating" question allowed multiple selections, which reflects the reality that procrastination is rarely caused by a single factor. In future iterations, adding a ranking element to this question could provide additional insight into which reasons respondents consider most significant.
- Response variation in open fields: A small number of respondents provided informal or non-English entries in open-ended fields, which reflects the diverse and candid nature of the sample. These responses were noted and the analysis focused on the structured options to maintain consistency.
- Self-reported performance ratings: Performance and work quality were self-assessed on a 1–5 scale. While self-perception is itself a valid and meaningful data point, pairing it with objective academic records in a future study could further strengthen the findings